Span XFRA — Investor & Strategic-Partner Report

Consolidated analysis of the distributed AI data center thesis, with a focused view on Northwest Ohio

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1. Executive summary

Span XFRA is the first credible attempt to convert latent residential electrical capacity into saleable AI inference capacity at hyperscale-grade hardware spec. Each node is a Dell PowerEdge box with 16× Nvidia RTX PRO 6000 Blackwell GPUs, 4× AMD EPYC CPUs, 3 TB of DDR5, direct-to-chip liquid cooling, and a ~15–16 kWh whole-home battery, drawing ~19.2 kW continuously from a host's 240 V residential service. Span pays the host's electric and internet bills (a flat ~$150/month, sometimes zero) and sells the compute to hyperscalers, neoclouds, and AI-native buyers. A 100-home, ~1.25 MW pilot ships in Q3 2026 with PulteGroup as the new-construction channel; Span has publicly floated 80,000 nodes nationwide and >1 GW of annual deployment capacity by 2027 **[S]**.

The thesis is investable because three structural conditions line up at once. Centralized data-center interconnection queues now run 5–7 years **[T]**; public opposition to greenfield hyperscale campuses is rising; and the political story Span tells utilities and PUCs — "*the distribution grid already runs at 40–45% utilization; we are putting compute on the headroom*" — is the most fundable narrative in DC infrastructure right now. The hardware is conventional and shippable. The Nvidia / Dell / Eaton / Pulte partnership stack legitimizes the product. The unit economics, on a derived basis, can clear. None of those facts are in serious dispute. What *is* in dispute, and what determines the size of the outcome, is whether (a) at least one public-utility commission approves a tariff that lets Span pay below retail for marginal kWh, (b) at least one hyperscaler signs a multi-year anchor contract at edge-premium pricing, and (c) the orchestration layer delivers hyperscaler-grade SLA across a residential fleet. The next 18 months — pilot results, first hyperscaler contract, first major PUC ruling — will resolve most of the load-bearing uncertainty.

Northwest Ohio is structurally the strongest US geography in which to play the distributed-compute thesis. Meta has already committed $800M and 350 MW to a Wood County campus that breaks ground in 2025 and goes online in 2027 **[T]** — confirmed hyperscale-tier demand in this exact footprint. AEP Ohio's data-center tariff (PUCO-approved July 2025) applies *only* to new loads above 25 MW aggregate and locks subscribers into 85% take-or-pay for up to 12 years **[T]**. XFRA nodes at ~19 kW are three orders of magnitude below the threshold and are billed on residential tariffs. The arbitrage door is real and currently open. AEP Ohio residential power runs ~9.94¢/kWh — below the NV/AZ benchmark of $0.13/kWh used in Span's own pilot economics. The result is a per-node gross margin band of roughly $128K (61%) in NW Ohio versus $113K (57%) in NV/AZ and $89K (45%) in California **[D]**. A 1,000-node regional pilot at base-case pricing implies ~$210M of annual gross revenue and ~$128M of regional gross margin before corporate overhead. Even the bear-case ($170K/node) clears $170M of regional revenue.

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2. The Northwest Ohio opportunity

2.1 Demand has already arrived

The single most important fact for any Northwest Ohio investment thesis is that **hyperscale-tier AI compute demand in this footprint is no longer theoretical.** Meta announced an $800M, 350 MW data center in Middleton Township, Wood County — roughly 20 miles south of Toledo — in April 2025. Ground broke in 2025; the campus targets online status in 2027. The JobsOhio / Regional Growth Partnership / Meta announcement is direct and confirmed; Meta lists it publicly as its 24th US data center and 28th globally **[T]**.

The Meta campus is not the only live prospect in the geography. Three other sites are in active motion as of late 2025:

- **Oregon, OH (Lucas County):** 170 acres sold November 2024 for $17.3M with up to eight buildings planned.
- **Waterville Township (Lucas County):** A developer eyeing 1,000–1,400 acres. The township imposed a 12-month moratorium on December 17, 2025.
- **Lucas County:** Commissioner Pete Gerken proposed a 90–120 day countywide pause in December 2025.

The political signal is unambiguous: hyperscale-scale builds are running into local opposition in exactly the same footprint where hyperscale demand is landing. That dynamic is *structurally favorable* for distributed and residential models that bypass the hyperscale-site permitting fight entirely. A 19 kW pod on a side yard is not the thing being moratoriumed in Waterville.

2.2 The AEP Ohio data-center tariff is an arbitrage door

On July 9, 2025 the Public Utilities Commission of Ohio approved AEP Ohio's data-center tariff; it became effective July 23, 2025. The tariff applies *only* to new loads above **25 MW aggregate** and imposes structural requirements that fundamentally reshape the economics for a hyperscale buyer **[T]**:

- **85% take-or-pay** of subscribed capacity for up to 12 years, regardless of actual usage.

- **4-year maximum** load ramp.
- **8-year minimum** contract term.
- **3-year minimum-charge** exit fee.

XFRA nodes are individually ~19 kW — three orders of magnitude below the 25 MW threshold — and are billed at residential rates. In the same regulatory geography, two compute footprints face wildly different cost structures: a hyperscaler pays for 85% of subscribed capacity for 12 years regardless of utilization; an aggregator of distributed residential nodes pays only for energy actually used. This is the single strongest defensible NW Ohio thesis point. It also explains why a hyperscaler signing the AEP tariff might *itself* want exposure to a distributed-compute layer in the same region as a portfolio hedge.

The standard ratepayer-advocate objection — that distributed compute "parks hyperscaler load on residential ratepayers" — is real and worth pricing in. Span's pre-emptive defense leans on the Brattle Group's *Untapped Grid* framing: that adding 7–12 kW of continuous controllable load behind the meter *increases* kWh sold over the same distribution capex base, so per-kWh delivery costs drop. Whether PUCO accepts that framing on its first contested filing is one of the watch-list items in §9.

2.3 Power here is actually cheap

Utility	Residential rate	Window
AEP Ohio	9.94 ¢/kWh	Jun 2025 – Aug 2026
Toledo Edison	~9.5–9.8 ¢/kWh	2025

Both rates are *below* the $0.13/kWh that Span's pilot economics assume for the Southwest. At ~19.2 kW continuous and ~134,500 kWh/yr per node, the per-node power line moves from ~$17.5K in NV/AZ to **~$13K in NW Ohio** — a $4.5K margin pickup before any other input changes **[D]**.

2.4 Per-node P&L: NW Ohio vs. comparison geographies

Line item	NW Ohio	NV/AZ (Span pilot)	California	Source
Power	**$13K**	$17.5K	$41K	Residential retail × 134,500 kWh/yr
Internet	$1K	$1K	$1K	$80/mo gigabit fiber
Operations	$7K	$7.5K	$8K	Truck rolls, monitoring, insurance
Capex amortization (5 yr)	$61K	$61K	$61K	$310K hardware ÷ 5 yr
Cost floor	**~$82K**	$87K	$111K	sum
Target revenue (base)	$210K	$200K	$200K	derived; see §5
Gross margin	**$128K (61%)**	$113K (57%)	$89K (45%)	

NW Ohio is structurally the strongest market in this comparison. The base case is not heroic: it assumes a \$2.50/GPU-hour wholesale price (mid-band of public retail rates discounted to wholesale), 85% inference utilization, and standard truck-roll cadence. The bull case (\$280K/node, ~\$198K margin) requires only a 1.5x edge-latency premium and PUCO blessing of residential aggregation. The bear case (\$170K/node, ~\$88K margin) assumes no edge premium and lower utilization — and still clears a \$170M regional revenue line at 1,000 nodes.

2.5 What the region brings beyond the numbers

Beyond raw economics, Northwest Ohio carries a small cluster of structural advantages that matter for execution speed:

- **First Solar's CdTe PV supply chain at Perrysburg**, giving Bet D (co-deployed solar) an unusual local anchor.
- **A workforce already absorbed by the Meta build.** IBEW Local 8 reports Journeyman Inside Wireman calls going unfilled daily as the Meta campus ramps; deploying distributed compute leverages the same trained pool.
- **UToledo PVIC and BGSU College of Engineering and Innovation** as technical validation partners.

The investor framing reduces to a single number: *\$210K/node base × 1,000 nodes = \$210M/yr regional gross revenue.* Even discounted to the bear case the order of magnitude clears the cost of running the six conversations needed to test it.

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3. What XFRA is

3.1 The company

Span (legal name SPAN.io, Inc.) was founded in 2018 in San Francisco by **Arch Rao**, who previously led Tesla Energy product. The company is best known for a smart electrical service panel that gives per-circuit visibility and dynamic load control. As of 2026 Span has raised >\$285M including a \$75M strategic investment from **Eaton** and is closing a Series C reported at ~\$176M (with \$163M of equity already sold as of February 2026) **[T]**.

XFRA ("eks-fra") was announced April 13, 2026 at Latitude Media's Transition-AI conference, with **Nvidia** as initial launch partner (silicon), **Dell** as server hardware provider, and **PulteGroup** — the third-largest US homebuilder — as the new-construction deployment channel. A white paper is published at xfra.ai. Span explicitly positions XFRA as *complementary* to centralized data centers: training stays in hyperscale campuses; inference and other latency-sensitive workloads (cloud gaming, autonomous-driving inference, medical-image diagnostics) move to the grid edge **[S]**.

3.2 The node

Component	Spec	Source
Chassis	Dell PowerEdge in outdoor enclosure (~HVAC condenser size)	Span; Network World
GPUs	16× Nvidia RTX PRO 6000 Blackwell Server Edition	Span; pv-magazine
CPUs	4× AMD EPYC	pv-magazine; Network World
Memory	3 TB DDR5	pv-magazine; Network World
Networking	24-port gigabit switch; fiber uplink	pv-magazine; Data Center Richness
Cooling	Direct-to-chip liquid; 35,000 BTU (~3-ton) heat pump	Technology.org; Data Center Richness
Acoustics	~60 dB operating	Data Center Richness; Network World
Battery	~15–16 kWh whole-home	Data Center Richness; Technology.org
Hardware BOM	>$250K of compute hardware per node	Network World analyst pricing

A derived check on price using street rates: RTX PRO 6000 ≈ $9–10K × 16 = $144–160K; EPYC ≈ $8.5–14K × 4 = $34–56K; 3 TB DDR5 DC-grade ≈ $90–105K; chassis, cooling, networking, enclosure, install ≈ $40–80K. **Total hardware ≈ $310–400K per node [D].** Network World's quoted ">$250K of compute hardware" is consistent with the lower-bound subset.

3.3 The power story

Span's claim: ~80 A continuous at 240 V ≈ **19.2 kW**, sized to fit inside a 200 A residential service envelope alongside the home's own loads **[S]**. A 16× Blackwell server at full load draws roughly 9.6 kW; 4× EPYC adds 1.0–1.6 kW; memory ~150 W; networking and chassis overhead ~200 W. IT load thus runs ~11–12 kW. Adding a 3-ton heat-pump cooling loop at ~3 kW electrical and battery-charging headroom at ~3–4 kW gives **17–19 kW total facility draw [D]** — at the top of the plausible band and consistent with Span's 19.2 kW figure.

A Latitude Media report describes the 100-home / 1,600-GPU pilot as "1.25 MW of compute capacity." That works out to 12.5 kW per node, which matches *IT-only* draw — not total facility draw. The 19.2 kW figure is the *electrical service allocation*. Both numbers are internally consistent if read carefully.

The political payload of the power story sits in one sentence from CEO Arch Rao: "*The existing distribution network operates at only 40 to 45% utilization, nominally.*" Span's panel measures real-time per-circuit draw, treats the XFRA pod as a managed always-on load, and curtails opportunistically when the host's appliances ramp. The included whole-home battery buffers short spikes, supplies the home during outages, and lets the pod participate in utility demand-response events instead of curtailing the homeowner. Span is in active conversations with utilities about "innovation in the tariff model" — i.e., the rate Span pays for XFRA-consumed kWh may diverge from standard residential retail **[S]**.

3.4 The homeowner offer

- Smart panel + XFRA pod + whole-home battery installed at **no upfront cost**.
- Flat **~$150/month** that covers the host's full electric *and* internet bill — Span pays both utilities directly.
- In the highest-value siting (cheap power + valuable inference latency + solar), the company has floated *free electricity and free internet*.
- The structural analog Span uses internally: a rooftop-solar PPA, except Span owns compute assets instead of panels.

3.5 Deployment trajectory

Phase	Date	Scope	Channel
In-house prototype	Pre-2026	Undisclosed paying customers	Direct
Proof of concept	Q3 2026	**100 new-construction homes / ~1.25 MW / ~1,600 GPUs** in NV or AZ	PulteGroup
Scale ramp	2027	>1 GW annual deployment capacity	Multiple homebuilders + utility partners
Commercial XFRA	Early 2027	Offices and small commercial buildings	TBD
Long-term stated goal	Unspecified	**~80,000 nodes nationwide**	Distributed

The new-construction-first choice is deliberate. Builder-channel deployment lets Span pre-wire the panel, conduit, battery pad, and liquid-cooling drain in one trip and pre-bundle the homeowner contract before move-in. Retrofit is technically possible but slower, more expensive, and runs into the older-residential-service envelope problem that the pilot deliberately sidesteps.

3.6 Headline unit-economics claim

- **Build cost:** ~$3M per MW of compute, vs. ~$15M/MW for a centralized 100 MW facility — roughly **1/5 the capex per MW [S]**.
- **Time-to-power:** **~6 months** to stand up the equivalent of a 100 MW facility (≈8,000 nodes), vs. 3–5 years for greenfield. This is the "speed-to-power" framing the marketing leans into.
- **Per-node asset-value benchmark:** Abe Yokell (Congruent Ventures) frames each node as ~$1M in asset value, derived from Nvidia's $50B/GW capex benchmark × 1 GW ÷ 50,000 nodes **[T]**.

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4. Technical viability

4.1 Hardware (low risk)

XFRA's bill of materials is a conventional enterprise AI inference server in an industrialized outdoor enclosure. Every component is shipping product; the integration is non-trivial but is not research-stage. Dell, Nvidia, and Eaton-grade enclosure design are all mature. Hardware risk is **low**.

4.2 Power envelope (low for new construction, high for retrofit)

The 19.2 kW claim is technically defensible for new-construction homes with a Span panel. For retrofit homes with older 100 A or 150 A service, the envelope is much tighter and may not work — which is why the pilot is in new-construction homes only. The retrofit question is deferred, not solved. It matters at the 5M-home scale; it does not matter at the 80,000-node scale.

4.3 Cooling and acoustic (low-to-medium risk)

Direct-to-chip liquid cooling is now standard for high-density AI racks. A 3-ton heat pump is a normal residential HVAC component; rejecting ~12 kW of IT heat is well within its design envelope. The 60 dB target is below most municipal residential nuisance ordinances (which typically set 55–65 dB nighttime limits at the property line), but Span has not published distance-attenuated numbers — the 60 dB figure is *at the unit*, not at the neighbor's window.

Risk flag. Long-term reliability of glycol/water loops in residential climates (freeze cycles, leaks, biological fouling) is the unsexy maintenance question. Glycol leak liability at a residential site is a real insurance question and Span has not publicly addressed it.

4.4 Networking (medium risk)

A 1 Gbps uplink is sufficient for inference traffic to and from one node but *insufficient* for large model-weight movement, KV-cache sharing across nodes, or any tight-coupling workload. This is consistent with Span's stated workload positioning (inference, not training). At national scale, fiber availability becomes a real screen on the addressable home base — Davies (Utah State) estimates a suitability ceiling of ~30–40% of US homes, partly driven by fiber.

4.5 Orchestration (most novel, least documented)

Span's claim: workloads are scheduled and routed across nodes based on latency requirements and available energy capacity; nodes drain when a host has an outage or panel headroom drops. The orchestration layer is the single most technically novel piece of the system. It is also the least publicly documented. Span has not disclosed latency targets between workload arrival and node selection; the partial-failure model (e.g., 200 nodes simultaneously lose internet during a regional outage); how buyer SLAs are translated into orchestrator policy; or whether GPU memory state is checkpointed and migrated or workloads simply restart on a new node.

These omissions are not red flags by themselves — companies don't publish their orchestrator design pre-launch. But they mean the orchestration story is **assertion, not evidence**, at this point. Risk grade:

medium-high.

4.6 Technical viability — summary

Component	Risk	Notes
Hardware	Low	Mature parts, conventional integration
Power envelope	Low (new) / High (retrofit)	Pilot avoids retrofit risk by selecting new-build only
Cooling	Low to medium	Standard tech; residential maintenance and insurance unproven
Acoustics	Low	Likely within ordinances; no distance-attenuated data
Networking	Medium	Constrains scale-out beyond fiber-pre-pulled footprint
Orchestration	Medium-high	Most novel; no public evidence yet

Verdict. Technical viability is plausible and Q3 2026-deliverable for the 100-home pilot. The unverified pieces only start to matter at the 1,000-to-10,000 node scale.

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5. Business viability and unit economics

5.1 The four-variable model

Span's pitch reduces to four numbers per node:

- **R** — annual compute revenue from selling node capacity to the buyer
- **C_pwr** — annual cost of paying the host's electricity bill
- **C_net** — annual cost of paying the host's internet bill
- **C_op** — annual amortized capex plus operating cost (truck rolls, software, support, insurance)

Per-node margin ≈ R − (C_pwr + C_net + C_op). The model only works if R is significantly larger than the sum.

5.2 Each term, derived

C_pwr. 19.2 kW × 8,760 hr × ~80% effective utilization ≈ 134,500 kWh/yr. At Nevada/Arizona residential retail of ~$0.13/kWh ≈ **$17,500/yr**. At California residential of ~$0.30/kWh ≈ $40,300/yr. **At AEP Ohio 9.94¢/kWh ≈ $13,000/yr.** The Southwest pilot picks the cheap end deliberately; Ohio is cheaper still.

C_net. Residential gigabit fiber ≈ $80/month × 12 = **~$960/yr**. Span pays the host's bill directly; some margin to the ISP, but small.

C_op. Unknown publicly. Truck roll to one rack at a residential site at $300–500 per visit, with 4–6 visits per year, ≈ $1,200–3,000/yr. Insurance, software, monitoring not published; estimated total **$5,000–10,000/yr**.

Capex amortization. \$310–400K hardware ÷ 5 years (aggressive but standard for GPU depreciation under AI-cycle pressure) ≈ **\$62,000–80,000/yr**.

Cost floor (Southwest): ~\$85,000–93,000/yr per node. **Cost floor (NW Ohio):** ~\$82,000/yr per node. **Cost floor (California):** ~\$108,000–116,000/yr per node.

5.3 The revenue side — three independent back-solves

The challenge is that Span has not published per-node revenue. Three independent back-solves converge in the same band.

Method A — retail GPU-hour pricing discounted to wholesale. Median retail rent for RTX PRO 6000 Blackwell on the open market is \$2.25–\$3.56 per GPU-hour as of May 2026 **[T]**. Span sells wholesale to hyperscalers, not retail to GPU renters — apply a standard wholesale-to-retail discount of 60–70%. Inference utilization typically lands at 80–90% (versus 60–70% for training):

Scenario	\$/GPU-hr	Utilization	Wholesale ratio	Annual/node
Bear	\$2.00	70%	0.55	\$108K
Base	\$2.50	85%	0.65	**\$193K**
Bull	\$3.00	90%	0.70	\$264K

Formula: \$/GPU-hr × 16 GPUs × 8,760 hrs × utilization × wholesale ratio.

Method B — Congruent Ventures asset-value back-solve. Yokell's \$1M asset-value-per-node framing, applied at a typical data-center asset yield (15–25% gross revenue / asset value), lands at **\$150K–\$250K/yr**.

Method C — Span's own framing. The CEO has stated revenue "exceeds the cost of the XFRA system itself" without disclosing a number. Span quotes \$3M/MW for XFRA capex vs. \$15M/MW for traditional builds; at 19.2 kW/node, \$3M/MW implies *system* capex of ~\$58K per node. If revenue exceeds that on a yearly basis, it is a hard floor — and Methods A and B both land well above it.

Convergence. The three methods cluster at **\$180K–\$230K per node per year**. The base case used throughout this report is **\$210K/yr**.

5.4 Where the model gets fragile

Sensitivity	Direction	Notes
GPU resale price compression	Negative	Blackwell-successor releases that cut inference cost-per-token by 50% drag R sharply
Residential-rate increases without offsetting tariff innovation	Negative	High-rate states get unprofitable fast
Hyperscaler reluctance to pay edge premium	Negative	If buyers prefer Crusoe's stranded-power compute at lower price
Channel slowdown (Pulte ramp)	Negative	Pulte ≈ 7–8% of ~1M annual US single-family completions
Single anchor hyperscaler signed at scale	Positive	Removes most R-side uncertainty
Inference-latency premium for consumer AI workloads materializes	Positive	AR/VR, agentic chat, autonomous-vehicle inference
PUC tariff innovation that lets Span pay below-retail for marginal kWh	Positive	The most leveraged single variable

5.5 Why a hyperscaler would pay the edge premium

The math requires hyperscaler-tier pricing roughly equivalent to $24–30 per GPU-hour fully-loaded **[D]**, against decentralized network prices of $1.50–$3.50/GPU-hr (io.net) and hyperscaler prices of $5–15/GPU-hr for similar-class GPUs. The premium has to be justified by one of:

- **Latency.** Inference within tens of milliseconds of the end user — a metro DC can match this (~5–20 ms to consumer ISPs); a Texas hyperscale campus cannot (~30–80 ms to East Coast users).
- **Availability.** Capacity that can be turned on in six months, not three to five years. The speed-to-power argument.
- **Renewable proximity / political optics.** Compute deployed at residential addresses generates a different ESG story than a 200 MW campus near a substation.

The first two are real and probably worth a premium. The third is soft.

5.6 Business viability — verdict

The economics can work at pilot scale in the Southwest and at modest scale in NW Ohio with conservative assumptions. They **almost certainly do not work** in California without tariff innovation. They **may or may not work** at 80,000-node national scale depending on inputs not currently public.

The story is not implausible. It is, however, *load-bearing on three things Span has not published*: orchestrator efficiency, anchor-tenant pricing, and utility tariff treatment.

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6. Competitive landscape

There is **no other company in the US putting a 16-Blackwell-GPU pod on the side of a single-family home.** In that exact niche, Span has the field to itself today. But XFRA is one answer to a much larger problem — *AI compute is bottlenecked by power and time-to-power, not by silicon* — and there are at least eight categories of solution chasing the same gap. They differ on where the compute lives, who pays for the power, what workload they target, and how they get around the grid.

6.1 Direct peers: distributed compute in homes and small buildings

Heata (UK). A ~1 kW compute server bolted to a domestic hot-water cylinder, dumping waste heat into the tank — up to ~4.25 kWh/day of hot water per host, saving up to £340/yr. Sells compute via Civo; works with British Gas. Offline batch only. Roughly two orders of magnitude smaller per node than XFRA.

Qarnot Computing (France). Founded 2010. Liquid-cooled servers feeding district-heating loops or building hot water (~95% heat reuse). CPU/HPC focus, French scale, ~$55M raised. The spiritual ancestor of "compute as a heat source." Not a serious threat.

Deep Green (UK). Immersion-cooled HPC tubs at UK leisure centres' swimming pools, ~28 kW per site. Backed by £200M from Octopus Energy. Closer to XFRA on cooling and per-site power, but B2B-to-leisure-centre, not residential, and general cloud workloads, not premium AI inference.

6.2 Modular / prefab data centers

The "speed-to-power middle ground" — same workloads as XFRA, very different unit size. Market valued at ~$29–33B in 2025.

- **Schneider + Compass:** prefabricated white-space pods, first units fall 2025; ~30% faster build, ~3 months sooner revenue.
- **Vertiv MegaMod CoolChip:** high-density liquid-cooled modular DC for AI; ~50% faster delivery than traditional.
- **Crusoe Spark:** 100s of kW to 100s of MW per unit, ~3 months time-to-power. New Spark Factory live in Brighton, CO; Energy Vault signed for up to 25 MW; Redwood Materials expanded from 4 to 24 Spark units.
- **Duos Edge AI:** 55 ft × 12.5 ft modular pods for edge AI inference.

These compete with Span on *time-to-power* but not on *power source* — they still need a real interconnection.

6.3 Edge colocation

EdgeConneX (North America, Europe, APAC; €500M expansion late 2024) and **EdgePresence** ($50M expansion 2024) own metro/regional edge sites and operate them as IaaS. They compete on the same latency story as XFRA but buy commercial power through the standard interconnection queue.

6.4 Hyperscaler and telco edge boxes

AWS Outposts, Azure Stack Edge, Google Distributed Cloud Edge, Verizon 5G Edge, AT&T edge, AWS Wavelength. Edge zones planted in telco facilities and customer premises. Outposts mindshare actually *declined* in April 2026 (9.3%, down from 15.4% YoY); Azure Stack Edge dropped from 28.9% to 14.6%. The appliance model is losing traction against hyperscaler edge zones and regional DCs.

The analyst Alex Cordovil (Dell'Oro), in his XFRA assessment, specifically flagged this category as "*the more instructive parallel*" to Span — telcos already have power, fiber, security, and a distributed-node structure, but "still wrestle with running compute across a small number of GPUs per site."

6.5 Behind-the-meter generation

If interconnection is the bottleneck, skip it. Generate on-site.

- **Crusoe Energy:** 1.2 GW Stargate campus in Abilene, TX, energized within ~1 year of breaking ground. Fast Company 2026 Most Innovative.
- **Bloom Energy + Brookfield:** $5B partnership to put SOFC capacity across AI factories worldwide. Bloom doubling annual manufacturing from 1 GW to 2 GW by end-2026; Q1 2026 revenue $751M, up 130% YoY. Bloom's 2026 Power Report: 32% of US data centers plan to operate fully off-grid on SOFC by 2030, up from 22% six months earlier.
- **FluxCore:** modular DC directly powered by on-site generation.

The category as a whole closed $7.65B in binding deals from late 2025 to early 2026 **[T]**.

6.6 Grid-flex orchestration

A different shape entirely: don't deploy new compute — make existing AI factories behave like flexible loads.

- **Emerald AI + Nvidia:** "Emerald Conductor" software ramps AI workloads in response to grid signals. Phoenix demo via EPRI's DCFlex: ramped down in 15 minutes, sustained a 25% reduction for 3 hours. Combined with Nvidia's DSX Flex, claims to unlock up to **100 GW** of new US grid capacity without building new plants **[T]**.
- **Verrus** (Sidewalk Infrastructure spinout): testing at NREL's 70 MW DC; shifts 100% of a DC's power demand from grid to on-site within one minute of a utility request.

Span and Emerald are pitching the same political story — grid utilization is the cheap unlock — from different angles. A hyperscaler could plausibly use both.

6.7 Decentralized GPU marketplaces

Software-only distribution layer over already-installed GPUs.

- **io.net:** $1.50–$3.50/GPU-hr; 15–63% discount on AWS-comparable configurations.
- **Akash Network:** launched AkashML for managed AI inference.
- **Render Network:** 600+ open-weight models on its AI Compute Subnet.

The decentralized GPU sector handled an estimated **$200M in annualized protocol revenue in early 2026 [T]**. Different product, same thesis. They threaten Span on the price-sensitive long tail, not on hyperscaler-grade inference.

6.8 Where Span actually wins, and where it doesn't

Span wins on:

- **Speed to incremental capacity in a specific metro.** Six months to stand up the equivalent of a 100 MW DC, no interconnection queue.
- **Latency proximity to consumers.** For consumer-facing inference (AI chat, gaming, on-device assistants), one residential hop from the end user is a real advantage telco edge can match but modular prefab cannot.
- **Capex per MW.** ~$3M/MW vs. ~$15M/MW greenfield.
- **Political story.** "Using grid headroom that's already there" is cleaner to a PUC than "building a 200 MW campus with a new substation."

Span loses or is at parity on:

- **Cluster topology for frontier training.** Modular prefab and BTM Crusoe beat XFRA on tightly-coupled workloads. Span concedes this.
- **Per-GPU-hour cost.** Decentralized networks beat enterprise pricing by 60–90% for workloads that tolerate weak SLA.
- **Operability and service economics.** A truck roll to one residential side yard is far more expensive than a tech walking between racks.
- **Physical-security trust.** Telco sites and BTM generation have decades of carrier-grade operating practice.
- **Capacity ceiling.** At 30–40% household suitability, Span's plausible US ceiling is single-digit GW.

Most direct structural threats, ranked:

1. **Telco edge (Verizon, AT&T, AWS Wavelength).** Same distributed shape, existing fiber + power + security. The most credible direct substitute.
2. **Crusoe Spark + Edge Zones.** Closest factory-built tiny-DC product, three-month time-to-power, commercial parcels rather than residences.
3. **Behind-the-meter + Emerald AI combo.** A hyperscaler that pairs Bloom SOFC with Emerald's grid-flex gets most of Span's speed-to-power benefit without residential channel friction.
4. **Modular prefab (Schneider+Compass, Vertiv).** Cuts greenfield to ~16 weeks.
5. **Heata / Qarnot / Deep Green.** Useful cohort, not a competitive constraint.
6. **Decentralized GPU networks.** Cost ceiling, not a strategic threat.

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7. Strategic positioning and risk inventory

7.1 The position Span is occupying

XFRA's position is "**first and only residential-channel distributed compute at hyperscaler-grade hardware spec.**" Three things define it:

1. **Channel exclusivity.** PulteGroup is the #3 US homebuilder; Span's relationship is the first builder-distributor for compute at this scale.
2. **Power thesis.** "Headroom on existing grid" is a politically distinct story from "we need to build a new substation" — defensible because it has utility and ratepayer alignment.
3. **Hardware credibility.** Nvidia + Dell + Eaton-grade design is a signal-rich partnership stack that legitimizes the product to hyperscaler buyers.

7.2 Moat analysis

Asset	Defensibility	Notes
Pulte partnership	Medium	Pulte can deploy with others; logo-strong but contractually unknown
Span panel installed base	High	5+ years of installs; real switching cost for builders already standardized on Span
Power-control IP and certification	High	Long process to UL-list and PUC-clear a panel with this behind-the-meter authority
Orchestration software	Low-medium	Pure software; replicable by any well-funded competitor
Hardware design	Low	Pure integration of commodity parts; no proprietary silicon
Tariff/regulatory relationships	Medium	Slow to build, slow to copy
Brand / first-mover narrative	Low	Decays as competitors enter

The **real moat** is the *combination* of panel installed base + builder channel + power-control IP. None of those alone is a moat; all three together are reasonably hard to copy in under 3–4 years.

7.3 Strategic timing

XFRA launches into a market where interconnection queues are 5–7 years and worsening; inference is becoming the dominant AI workload; public opposition to centralized DCs is rising; behind-the-meter generation is having its capital moment ($7.65B in fuel-cell deals Q4 2025–Q1 2026); and utilities are actively exploring tariff and grid-flex programs (EPRI DCFlex, NREL, Verrus, Emerald AI). The timing is good. If Span had launched two years earlier, the headroom story would have landed in a less-receptive market.

Position is defensible for **3–4 years** on the unique combination of installed base + builder channel + regulatory work. Beyond that, defensibility depends on whether the orchestration layer becomes a meaningful product differentiator (low confidence) or whether anchor-tenant contracts and tariff treatments lock in distribution (medium confidence). The position is **fragile to two things:** anchor-tenant non-renewal (if a hyperscaler walks, economics force consolidation) and PUC reversal (if a major state rules against residential headroom resale, the model loses its primary market).

7.4 Risk inventory

Technical:

Risk	Severity	Likelihood
Glycol leak at residential site	Medium	Low
Cooling system failure in extreme heat (Phoenix August)	Medium	Medium
Orchestrator partial-failure cascade	High	Low-medium
GPU thermal throttling under residential cooling envelope	Medium	Medium
Tampering / physical-security breach	High	Low at pilot, rising at national

Operational:

Risk	Severity	Likelihood
Truck-roll cost balloons as fleet grows	High	High
Host churn (homeowner sells, divorces)	Medium	Medium
Internet provider non-cooperation	Low	Low
Recall or rolling firmware issue	High	Low

Market / business:

Risk	Severity	Likelihood
Hyperscaler walks from anchor contract	Critical	Low at pilot, medium at scale
GPU resale price compression	High	Medium-high
Latency premium fails to materialize	High	Medium
Decentralized GPU networks undercut	Medium	High but different workload
Channel slowdown (Pulte ramp)	Medium	Low-medium

Regulatory / political:

Risk	Severity	Likelihood
PUC denies favorable tariff for XFRA load	Critical	Medium
Ratepayer-advocate challenges model	High	Medium-high
HOA / local zoning ban for "data center on residential lot"	Medium	Medium
Federal regulation of distributed compute as critical infrastructure	Low-medium	Low
Tax-treatment changes	Medium	Low

Catastrophic / tail:

Risk	Severity	Likelihood
House fire attributed to XFRA unit	Critical	Very low
Mass GPU theft (organized)	High	Low at pilot, rising
Cybersecurity incident exposing buyer data	Critical	Low
Climate event taking out a regional cluster	Medium	Climate-dependent

The **two binary risks** that dominate the thesis are PUC tariff outcomes (existential, plausible) and truck-roll cost scaling (probable, structural). The most over-discussed risks in press coverage are acoustic/visual neighborhood objections (real but manageable via the HVAC form factor) and NIMBY (which the pilot's new-construction-only design largely sidesteps).

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8. Adjacent opportunities and four directional bets

8.1 Adjacencies Span isn't currently pursuing

Eight extensions are visible from the analysis. Listed in order of strategic distance from XFRA's current scope:

1. **Bundled grid-services revenue.** Span has the panel, orchestration, and battery. Selling demand-response capacity to utilities as a separate revenue stream alongside compute resale is a natural extension at no incremental capex.
2. **Heat reuse for residential hot water.** The 12 kW IT load is rejected as low-grade heat. In northern climates, capturing it to pre-warm domestic hot water (Heata-style) could reduce a host's heating bill 30–50% in winter and turn "free electricity" into "free electricity and free hot water."
3. **Sovereign / institutional channel.** The same hardware in the same enclosure at a school, fire station, library, or municipal building serves a different buyer (the institution itself) at different economics (host-as-customer). Sidesteps PUC risk entirely.
4. **Insurance-as-a-product.** The glycol/cooling/fire risk at residential addresses creates a real insurance product opportunity. At fleet scale, a 10–20% margin lift; as a side effect, makes the host conversation easier.

5. **Hardware-as-a-service to non-Span markets.** The XFRA enclosure (HVAC-form-factor outdoor liquid-cooled GPU pod) sold as a product to buyers who don't want the full service model — telcos for cell-tower compute, retailers, smaller builders, federal facilities.
6. **Multi-rack site upgrade path.** If one host site proves out at 19 kW, the same lot with a slightly larger pad can host 2–4 racks. Unlocks tightly-coupled workloads at sites where the host is willing — likely commercial, civic, or large-rural-residential.
7. **Geographic expansion to high-heating-cost markets.** The Southwest is easiest to *start*; the most economically interesting US market for the residential-headroom thesis combined with heat reuse is the **Northeast**, where residential rates are high but heating costs are higher.
8. **Federal / DOD edge inference.** The same product, contracted federally for on-base deployment, sidesteps the entire PUC question and likely fetches much higher per-node revenue. Not currently pursued.

Of these, (1) utility grid services and (6) multi-rack upgrade are nearest to Span's current capability. (3) sovereign, (7) Northeast with heat reuse, and (8) federal are larger strategic shifts but reuse most of the existing investment.

8.2 Four directional bets: "what if it isn't Span"

If a different operator wanted to build something in the same neighborhood as XFRA — small, distributed compute solving the speed-to-power gap — but deliberately chose a different shape, there are four coherent archetypes worth naming. Each is a *bet*, not a recommendation. Each one beats Span on something specific and loses to Span on something specific.

Bet A — "Hearth": heat reuse for the US Northeast. The first US Heata. Compute heats hot water in homes that currently pay $0.30+/kWh for heat. Compute is incidental; the value is reclaimed heat. Workload is batch / offline AI training and rendering. Channel is utility energy-efficiency programs + insulation/HVAC installers. Hardware is 1U-class (~1 kW) on the hot-water tank. Economic model is heat-reclamation savings for the host + compute resale. Geography: New England + NY.

Why it works. Northeast heating costs are genuinely expensive ($2,000–$4,000/yr per home is normal). Reclaiming that as a free byproduct of compute is a consumer-grade pitch. Doesn't compete with Span — serves a workload Span can't take.

Why it might not. Single-box compute is hard to differentiate. Utility EE programs are slow. Batch market is small and price-pressured.

Bet B — "Steeple": tiny DCs at cell towers and telco huts. Take Span's pitch and put it on infrastructure already built to host equipment, with commercial power, fiber, security, and a tenant relationship. Channel: tower co-owners (American Tower, Crown Castle) + WISPs in rural. Hardware envelope: 2–4 racks per site (~40–80 kW). Workload: low-latency inference (consumer AI, AR/VR, autonomous driving).

Why it works. This is exactly the parallel Cordovil pointed to in his XFRA assessment. Towers/huts have power, security, fiber, and operators who already do site rounds. Per-site economics are way better than residential. No ratepayer politics.

Why it might not. This is the obvious move and AWS Wavelength / Verizon 5G Edge are already on it. Tower co-owners want a cut. Buyer doesn't care where the box is if both meet latency.

Bet C — "Civic": sovereign inference at municipal/county facilities. Sell a turnkey "local AI compute" box to schools, libraries, fire stations, town halls, hospitals, regional government. The selling point isn't speed-to-power — it's data sovereignty and local control. Hardware: 1–2 racks per site. Economic model: annual contract (host-as-customer) with optional shared-capacity overflow.

Why it works. US schools and small munis can't use most AI offerings because of data governance (COPPA, FERPA, HIPAA), IT staffing, and political sensitivity. A local box is a real answer. Government contracts are sticky. Nobody else is targeting this buyer directly.

Why it might not. Government procurement is glacially slow. School IT budgets are small. Margin per site is low. "AI in schools" is politically volatile.

Bet D — "Confluence": tiny DC co-deployed with new utility-scale solar+storage. Solar developers already have to build grid interconnections that are partly underused most of the day. Co-locating a tiny DC on the parcel uses that underutilization, gives the developer extra revenue, and gives the compute buyer cheap renewable-tagged inference. Channel: NextEra, Invenergy, AES, EPCs. Hardware: 2–4 racks per parcel scaling to 5–10. Geography: solar-rich states with long interconnection queues (TX, AZ, NV, CA).

Why it works. Solar developers have a real problem (interconnection queues, curtailment) compute can absorb. Buyers want renewable-tagged inference for ESG. The compute is genuinely cheaper because it's stranded power. No residential politics.

Why it might not. Solar developers are slow procurement targets. The renewable-tagged premium may collapse as RECs and PPAs commoditize. Crusoe is likely to move here.

Trade-off summary.

Bet	Channel	Hardware	Workload	Beats Span on	Loses to Span on
Hearth (A)	Utility EE + HVAC installers	1U / ~1 kW	Batch	Northeast economics; consumer-grade pitch	Per-site capacity; not competing for inference dollars
Steeple (B)	Tower co-owners + WISPs	2–4 racks / ~40–80 kW	Low-latency inference	Per-site ops; existing power & fiber	Residential proximity for very-last-mile latency
Civic (C)	Direct gov procurement	1–2 racks / ~20–40 kW	Sovereign inference	Sticky revenue; no PUC risk	Sales cycle; ceiling on price per site
Confluence (D)	Solar developers + EPCs	2–4 racks scaling to 5–10	Inference + scheduled training	Capex per MW; renewable tag	Speed to first site (depends on solar cadence)

For Northwest Ohio specifically, **Bet D (Confluence) is the unusually strong local fit** because of First Solar's CdTe PV supply chain at Perrysburg and the workforce already trained for the Meta build. Bet C (Civic) is the second-strongest fit, given the density of UToledo / BGSU institutional buyers in the geography.

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9. Watchlist and decision gates

9.1 What we still don't know — priority order

In order of how much each, if answered, would change the assessment:

1. **First-tenant pricing and contract terms.** Who buys, how much, for how long, at what price.
2. **First PUC ruling on XFRA-relevant tariff.** Likely Nevada or Arizona — sets precedent for high-rate states and for OH ratepayer-advocate posture.
3. **Per-node operating cost from the pilot.** Truck-roll frequency, glycol/cooling maintenance events, failure rates.
4. **Orchestrator architecture and partial-failure behavior.** Public technical disclosure or independent audit.
5. **Host satisfaction and churn at month 12 of the pilot.** The home-as-host model is novel; churn data is the leading indicator of channel scaling.
6. **Commercial XFRA spec and pricing (announced for early 2027).** Determines whether Span is residential-only or a broader distributed- compute brand.
7. **Insurance carrier underwriting position on residential compute pods.** The first major carrier to write or refuse coverage sets the market.
8. **Hyperscaler edge-strategy disclosures (AWS, Azure, Google).** If the hyperscalers commit publicly to internal edge buildouts, XFRA's addressable market shrinks.

9.2 What an investor would want before writing a check

1. **Anchor-tenant LOI.** Span has not publicly disclosed a per-node price. The whole valuation hinges on a hyperscaler signing wholesale terms. Until one does, the $210K target is derived, not market-validated.
2. **PUCO posture on distributed-compute aggregation.** AEP's >25 MW tariff is the arbitrage. Ratepayer advocates are already framing distributed compute as "hyperscalers parking load on residential ratepayers." A future PUCO ruling could close the door.
3. **AEP-Dayton zone-specific wholesale prices.** PJM-wide RT LMP averaged $51/MWh in 2025 (+47% YoY). AEP-Dayton-specific data is harder to pull; if it runs >$60/MWh, behind-the-meter residential-resale economics tighten. Flag as thin data.
4. **The $3M/MW vs. $310K-hardware gap.** Span's published *system* cost of $3M/MW implies $58K/node — but a 16× RTX PRO 6000 Blackwell server alone is $200K+ at street prices. Either the figure excludes GPUs (most likely — GPUs depreciate on a different cycle), or there's significant subsidy or financing in the mix. Worth a direct question to Span.
5. **Truck-roll cost at scale.** Ops at $7K/yr is an estimate; actual data from the 100-home pilot won't surface until Q3 2026.

9.3 Decision gates

Two explicit gates an investor should treat as binary on/off:

- **Gate 1 — Anchor tenant.** A signed multi-year hyperscaler contract at edge-premium pricing. Without it, the revenue side of the model is derived, not validated. Likely resolution window: Q4 2026 – Q2

2027.

- **Gate 2 — First major PUC ruling.** A favorable ruling on residential aggregation in any of NV, AZ, or OH would de-risk the model nationally; an adverse ruling in any high-rate state would put the model on defense. Likely resolution window: 2027.

A "yes" on both converts XFRA into a category-defining business with a 3–4 year structural moat. A "no" on either likely confines XFRA to a profitable-but-bounded business in low-rate markets — a real outcome, but not the gigawatt-scale ambition publicly framed.

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